

Peter Afif

714-512-0070 | peteraff19@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

CS graduate (CSUF, July 2026) seeking full-stack or backend engineering roles — open to remote or relocation.
Shipped 3 deployed apps with React, TypeScript, Node.js, PostgreSQL, Docker, and GitHub Actions CI/CD.

EDUCATION

California State University, Fullerton

Bachelor of Science in Computer Science, GPA: 3.2

Fullerton, CA

Aug. 2024 – July 2026

Orange Coast College

Associate's in Computer Science, GPA: 3.2

Costa Mesa, CA

Aug. 2021 – May 2024

PROJECTS

ResumeShift ↗ | Full-Stack | React, Node.js, Express, GPT-4o, OpenAI API | Deployed

- Built an AI-powered resume analyzer using **GPT-4o** across a 5-phase stateful **multi-turn conversation** — goal extraction, JD grounding, line-level analysis, bullet rewrites, and adjacent role matching — with a **React/Vite** frontend and **Node.js/Express** backend
- Designed a custom **eval framework** measuring **keyword hit rate** across 4 prompt iterations; improved score from **0.10** → **0.81** through targeted prompt changes and eval methodology fixes across 10 labeled test cases
- Engineered a **GPT-4o-mini** classifier to detect phase completion from free-text input, replacing brittle regex and enabling automatic **phase chaining** server-side; implemented optional **JD grounding** to extract role-specific competency phrases absent from the resume

Groove Vault: Full-Stack Practice Tracker ↗ | Full-Stack | React, TypeScript, Node.js, PostgreSQL, GitHub Actions

- Engineered a full-stack **TypeScript** web app (**React**, **Express.js**, **PostgreSQL**) with **JWT** + **Google OAuth** auth, real-time **Web Audio API** metronome, and BPM progression analytics; deployed to Render
- Built a **RESTful API** with **Express.js** using a **window function CTE** (**ROW_NUMBER()** **OVER PARTITION BY**) on **PostgreSQL** to compute per-rudiment BPM trend deltas across sessions
- Configured a **CI/CD** pipeline with 5 parallel **GitHub Actions** jobs (type-check, lint, build, **Vitest** unit tests, **Supertest** integration tests) against a fully mocked database

Backbeat AI ↗ | Machine Learning | Python, Scikit-learn, Librosa, React, Node.js, PostgreSQL, Docker

- Engineered a **feature engineering** pipeline extracting 16 rhythm features with **Python** and **Librosa** (IOI stats, beat deviation, tempo delta) after diagnosing that **Wav2Vec2** embeddings were blind to ms-level timing; improved model accuracy from 26% to 44% (1.76× above random baseline) across 5 iterations
- Built a data augmentation pipeline generating ~57k balanced training samples from 503 **Groove MIDI Dataset** files using **FluidSynth** and MIDI **set_tempo** ramps to simulate realistic rushing and dragging across a stratified train/val/test split
- Trained a **GradientBoostingClassifier** (**Scikit-learn**) with balanced class weights, improving dragging recall from 0.16 to 0.45 and rushed recall from 0.39 to 0.54; containerized backend via **Docker** on Render and deployed React frontend to **Vercel** with real-time **WebSocket** analysis and a live session dashboard

EXPERIENCE

Music Instructor

McCormick Music, Kidz Rock, Drum Path

2019 – Present

Orange County, CA

- Designed and delivered structured curricula for **50+ students** across private and group settings, tracking individual progress and adapting lesson plans based on measurable performance data

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL, HTML, CSS

Frameworks: React, Node.js, Express.js, Tailwind CSS, Vite

Databases: PostgreSQL

Developer Tools: Git, GitHub Actions, Docker, Linux

Libraries: Scikit-learn, Librosa, NumPy, OpenAI API, JWT, Vitest, Supertest

Concepts: Machine Learning, Data Structures & Algorithms, Object-Oriented Programming, RESTful APIs